

TIGAR 50

(Tigecycline Powder for Solution for Infusion 50mg/Vial, 5 mL Lyo Vial)

Description and Composition:

Tigecycline is a yellow to orange colored hygroscopic powder. TIGAR 50 (Tigecycline powder for solution for infusion 50mg/Vial, 5ml Lyo Vial) is sterile, lyophilized cake/ powder for Injection filled in 5mL USP Type I clear tubular glass Lyo Vial, administered by I.V or I.V infusion. Lactose Monohydrate is used as Bulking Agent, Hydrochloric Acid is used as Acidifying agent, Sodium Hydroxide used as Alkalizing Agent and Water for Injection as solvent.

Description after Reconstitution:

TIGAR 50 (Tigecycline powder for solution for infusion 50mg/Vial, 5ml Lyo Vial) is diluted with 5.3 mL of 0.9% Sodium Chloride or 5% Dextrose Injection to achieve a concentration of 10mg/mL of Tigecycline.

Dilution in Sodium Chloride Injection (0.9% w/v) – yellow to orange color solution.

Dilution in Dextrose Injection (5.0% w/v) - yellow to orange color solution.

Dilution in Lactated Ringer's solution - yellow to orange color solution.

Pharmacodynamics

Pharmacotherapeutic group: Antibacterial for systemic use, Tetracyclines, ATC code: J01AA12.

Mechanism of action

Tigecycline, a glycylcycline antibiotic, inhibits protein translation in bacteria by binding to the 30S ribosomal subunit and blocking entry of amino-acyl tRNA molecules into the A site of the ribosome. This prevents incorporation of amino acid residues into elongating peptide chains.

Tigecycline carries a glycyamido moiety attached to the 9-position of minocycline. The substitution pattern is not present in any naturally occurring or semisynthetic tetracycline and imparts certain microbiologic properties to tigecycline.

In addition, tigecycline is able to overcome the two major tetracycline resistance mechanisms, ribosomal protection and efflux. Accordingly, tigecycline has activity against a broad spectrum of bacterial pathogens. There has been no cross resistance observed between tigecycline and other antibiotics. No antagonism between tigecycline and other commonly used antibiotics. In general, tigecycline is considered bacteriostatic; however, TIGAR 50 has demonstrated bactericidal activity against isolates of *S. pneumoniae* and *L. pneumophila*.

Mechanism(s) of Resistance

To date there has been no cross-resistance observed between tigecycline and other antibacterial. Tigecycline is not affected by the two major tetracycline-resistance mechanisms, ribosomal protection and efflux. Additionally Tigecycline is not affected by resistance mechanisms such as beta-lactamases (including extended spectrum beta- lactamases), target-site modifications, macrolide efflux pumps or enzyme target changes (e.g. gyrase/topoisomerases). Tigecycline resistance in some bacteria (e.g. *Acinetobacter calcoaceticus-Acinetobacter baumannii complex*) is associated with multi-drug resistant (MDR) efflux pumps.

Interaction with Other Antimicrobials

In vitro studies have not demonstrated antagonism between tigecycline and other commonly used antibacterial.

Tigecycline has been shown to be active against most of the following bacteria, both in vitro and in clinical infections.

Aerobic and Facultative Gram- Positive Microorganism

Enterococcus faecalis (vancomycin-susceptible isolates)

Staphylococcus aureus (methicillin-susceptible and -resistant isolates)

Streptococcus agalactiae

Streptococcus anginosus grp. (Includes *S. anginosus*, *S. intermedius*, and *S. constellatus*)

Streptococcus pyogenes

Aerobic and Facultative Gram – Negative

Citrobacter freundii

Enterobacter cloacae

Escherichia coli

Klebsiella oxytoca

Klebsiella pneumoniae

Anaerobic Microorganism

Bacteroides fragilis

Bacteroides thetaiotaomicron

Bacteroides uniformis

Bacteroides vulgatus

Clostridium perfringens

Peptostreptococcus micros

The following bacteria exhibit minimum inhibitory concentrations (MICs) that are at concentrations that are achievable using the prescribed dosing regimens. However, the clinical significance of this is unknown because the safety and effectiveness of tigecycline in treating clinical infections due to these bacteria have not been established in adequate and well-controlled clinical trials.

Aerobic and Facultative Gram – Positive Microorganism

Enterococcus avium

Enterococcus casseliflavus

Enterococcus faecalis (vancomycin-resistant isolates)

Enterococcus faecium (vancomycin-susceptible and -resistant isolates)

Enterococcus gallinarum

Listeria monocytogenes

Staphylococcus epidermidis (methicillin-susceptible and -resistant isolates)

Staphylococcus haemolyticus

Aerobic and Facultative Gram – Negative Microorganism

Acinetobacter baumannii

Aeromonas hydrophila

Citrobacter koseri

Enterobacter aerogenes

Pasteurella multocida

Serratia marcescens

Stenotrophomonas maltophilia

Anaerobic Microorganism

Bacteroides distasonis

Bacteroides ovatus

Peptostreptococcus sp.

Porphyromonas sp.

Prevotella sp.

Other bacteria

Mycobacterium abscessus

Mycobacterium chelonae

Mycobacterium fortuitum

The development of tigecycline resistance in *Acinetobacter* infections seen during the course of standard treatment. Such resistance appears to be attributable to an MDR efflux pump mechanism. While monitoring for relapse of infection is important for all infected patients, more frequent monitoring in this case is suggested. If relapse is suspected, blood and other specimens should be obtained and cultured for the presence of bacteria. All bacterial isolates should be identified and tested for susceptibility to tigecycline and other appropriate antimicrobials.

Pharmacokinetics

Absorption

Tigecycline is administered intravenously and therefore has 100 % bioavailability.

Distribution

The plasma protein binding of tigecycline ranges from approximately 71 % to 89 % at concentrations observed in clinical studies (0.1 to 1.0 µg/ml) and the steady-state volume of distribution of tigecycline averaged 500

to 700 L (7 to 9 L/kg), indicating that tigecycline is extensively distributed beyond the plasma volume and concentrates into tissues.

Biotransformation

On average, it is estimated that less than 20 % of tigecycline is metabolized before excretion.

Elimination

The total clearance of tigecycline is 24 L/h after intravenous infusion. Renal clearance is approximately 13 % of total clearance. Tigecycline shows a poly exponential elimination from serum with a mean terminal elimination half-life after multiple doses of 42 hours although high inter individual variability exists.

Special populations

Patient with Hepatic Insufficiency

The single-dose pharmacokinetic disposition of tigecycline was not altered in patients with mild hepatic impairment. However, systemic clearance of tigecycline was reduced by 25 % and 55 % and the half-life of tigecycline was prolonged by 23 % and 43 % in patients with moderate or severe hepatic impairment (Child Pugh B and C), respectively.

Patient with Renal Insufficiency

The single dose pharmacokinetic disposition of tigecycline was not altered in patients with renal insufficiency (creatinine clearance <30 ml/min, n=6). In severe renal impairment, AUC was 30 % higher than in subjects with normal renal function.

Patient with Elderly Patients

No overall differences in pharmacokinetics were observed between healthy elderly subjects and younger subjects.

Patient with Paediatric Population

The safety and efficacy of tigecycline in the paediatric population 8 to <18 years of age have not been established.

Gender

There were no clinically relevant differences in the clearance of tigecycline between men and women. AUC was estimated to be 20 % higher in females than in males.

Race

There were no differences in the clearance of tigecycline based on race.

Weight

Clearance, weight-normalised clearance, and AUC were not appreciably different among patients with different body weights, including those weighing ≥125 kg. AUC was 24 % lower in patients weighing ≥125 kg. No data is available for patients weighing 140 kg and more.

Indications

TIGAR 50 is a tetracycline-class antibacterial indicated for the treatment of infections caused by susceptible isolates of the designated microorganisms in the conditions listed below for patients 18 years of age and older:

Complicated Skin and Skin Structure Infections

Complicated skin and skin structure infections caused by *Escherichia coli*, *Enterococcus faecalis* (vancomycin-susceptible isolates), *Staphylococcus aureus* (methicillin-susceptible and -resistant isolates), *Streptococcus agalactiae*, *Streptococcus anginosus* grp. (Includes *S. anginosus*, *S. intermedius*, and *S. constellatus*), *Streptococcus pyogenes*, *Enterobacter cloacae*, *Klebsiella pneumoniae*, and *Bacteroides fragilis*.

Complicated Intra-abdominal Infections

Complicated intra-abdominal infections caused by *Citrobacter freundii*, *Enterobacter cloacae*, *Escherichia coli*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Enterococcus faecalis* (vancomycin-susceptible isolates), *Staphylococcus aureus* (methicillin-susceptible and -resistant isolates), *Streptococcus anginosus* grp. (includes *S. anginosus*, *S. intermedius*, and *S. constellatus*), *Bacteroides fragilis*, *Bacteroides thetaiotaomicron*, *Bacteroides uniformis*, *Bacteroides vulgatus*, *Clostridium perfringens*, and *Peptostreptococcus micros*.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of TIGAR 50 and other antibacterial drugs, TIGAR 50 should be used only to treat or prevent infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

Appropriate specimens for bacteriological examination should be obtained in order to isolate and identify the causative organisms and to determine their susceptibility to tigecycline. TIGAR 50 may be initiated as empiric monotherapy before results of these tests are known.

Recommended dose

The recommended dosage regimen for TIGAR 50 is an initial dose of 100 mg, followed by 50 mg every 12 hours. Intravenous infusions of TIGAR 50 should be administered over approximately 30 to 60 minutes every 12 hours.

The recommended duration of treatment with TIGAR 50 for complicated skin and skin structure infections or for complicated intra-abdominal infections is 5 to 14 days. The duration of therapy should be guided by the severity and site of the infection and the patient's clinical and bacteriological progress.

Use in patients with renal impairment

No dosage adjustment of tigecycline is necessary in patients with renal impairment or in patients undergoing hemodialysis.

Use in patients with hepatic impairment

No dosage adjustment is necessary in patients with mild to moderate hepatic impairment (Child Pugh A and Child Pugh B). Based on the pharmacokinetic profile of tigecycline in patients with severe hepatic impairment (Child Pugh C), the dose of tigecycline should be altered to 100 mg followed by 25 mg every 12 hours. Patients with severe hepatic impairment (Child Pugh C) should be treated with caution and monitored for treatment response.

Use in children

Safety and effectiveness in patients under 18 years of age have not been established. Therefore, use in patients under 18 years of age is not recommended.

Use in elderly

No dosage adjustment is necessary in elderly patients.

Race and gender

No dosage adjustment is necessary based on race or gender.

Preparation and Handling

Each vial of TIGAR 50 should be reconstituted with 5.3 mL of 0.9% Sodium Chloride Injection, USP or 5% Dextrose Injection, USP or Lactated Ringers Solution to achieve a concentration of 10 mg/mL of tigecycline. (Note: Each vial contains a 6% overage, thus, 5mL of reconstituted solution is equivalent to 50mg of the drug.) The vial should be gently swirled until the drug dissolves. Withdraw 5 mL of the reconstituted solution from the vial and add to a 100 mL intravenous bag for infusion (for a 100 mg dose, reconstitute two vials; for a 50 mg dose, reconstitute one vial). The maximum concentration in the intravenous bag should be 1 mg/ mL. The reconstituted solution should be yellow to orange in colour; if not, the solution should be discarded. Parenteral drug products should be inspected visually for particulate matter and discoloration (e.g., green or black) prior to administration.

Once reconstituted, TIGAR 50 mixed with 0.9% Sodium Chloride Injection, USP or 5% Dextrose Injection, USP or Lactated Ringers Solution, USP may be stored at 25°C for up to 24 hours (6 hours in the vial and the remaining time in the intravenous bag). If the storage conditions exceeds 25°C after reconstitution, tigecycline should be used immediately. Alternatively, TIGAR 50 mixed with 0.9% Sodium Chloride Injection, USP or 5% Dextrose Injection,, USP may be stored refrigerated at 2 °C to 8 °C for up to 48 hours following immediate transfer of the reconstituted solution into the intravenous bag.

TIGAR 50 may be administered intravenously through a dedicated line or through a Y- site. If the same intravenous line is used for sequential infusion of several drugs, the line should be flushed before and after infusion of TIGAR 50 with 0.9% Sodium Chloride Injection, or 5% Dextrose Injection or Lactated Ringers Solution. Injection should be made with an infusion solution compatible with tigecycline and with any other drug(s) administered via this common line.

Compatibilities

Compatible intravenous solutions include sodium chloride 9 mg/ml (0.9 %) solution for injection, dextrose

50 mg/ml (5 %) solution for injection, and Lactated Ringer’s solution for injection. When administered through a Y-site, TIGAR 50 is compatible with the following drugs or diluents when used with either 0.9% Sodium Chloride Injection, USP or 5% Dextrose Injection, USP: amikacin, dobutamine, dopamine HCl, gentamicin, haloperidol, Lactated Ringer’s, lidocaine HCl, metoclopramide, morphine, norepinephrine, piperacillin/tazobactam (EDTA formulation), potassium chloride, propofol, ranitidine HCl, theophylline, and tobramycin.

Incompatibilities

The following drugs should not be administered simultaneously through the same Y-site as TIGAR: amphotericin B, amphotericin B lipid complex, diazepam, esomeprazole and omeprazole.

Mode of administration

Intravenous Infusion.

Contraindication

TIGAR 50 is contraindicated for use in patients who have known hypersensitivity to tigecycline.

Warning and precautions

Tigecycline may be associated with permanent tooth discoloration in the teeth in humans during tooth development.”

Pseudomembranous colitis has been reported with all nearly antibacterial agents and may range in severity from mild to life-threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of any antibacterial agent.

Anaphylaxis/Anaphylactoid Reactions

Anaphylaxis/anaphylactoid reactions have been reported with nearly all antibacterial agents, including TIGAR 50, and may be life-threatening. TIGAR 50 is structurally similar to tetracycline-class antibiotics and should be administered with caution in patients with known hypersensitivity to tetracycline-class antibiotics.

Clostridium difficile Associated Diarrhoea

Clostridium difficile associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents, including TIGAR 50, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

If CDAD is suspected or confirmed, ongoing antibiotic use not directed against *C.difficile* may need to be discontinued. Appropriate fluid and electrolyte management, protein supplementation, antibiotic treatment of *C. difficile*, and surgical evaluation should be instituted as clinically indicated.

Precautions

Tetracycline-Class Effects

Glycylcycline class antibiotics are structurally similar to tetracycline class antibiotics and may have similar adverse effects. Such effects may include: photosensitivity, pseudotumor cerebri, and anti-anabolic action (which has led to increased BUN, azotemia, acidosis, and hyperphosphatemia). As with tetracycline, pancreatitis has been reported with the use of TIGAR 50.

Superinfection

As with other antibacterial drugs, use of TIGAR 50 may result in overgrowth of non- susceptible organisms, including fungi. Patients should be carefully monitored during therapy. If superinfection occurs, appropriate measures should be taken.

Interactions with other medicaments

Concomitant administration of tigecycline and warfarin (25 mg single-dose) to healthy subjects resulted in a decrease in clearance of R-warfarin and S-warfarin by 40 % and 23%, and an increase in AUC by 68 % and 29 %, respectively. The mechanism of this interaction is still not elucidated. Available data does not suggest that this interaction may result in significant INR changes. However, since tigecycline may prolong both prothrombin time (PT) and activated partial thromboplastin time (aPTT), the relevant coagulation tests should be closely monitored when tigecycline is co-administered with anticoagulants .Warfarin did not affect the pharmacokinetic profile of tigecycline. Tigecycline is not extensively metabolized. Therefore, clearance of tigecycline is not expected to be affected by active substances that inhibit or induce the activity of the CYP450 isoforms. *In vitro*, tigecycline is neither a competitive inhibitor nor an irreversible inhibitor of CYP450 enzymes.

Tigecycline in recommended dosage did not affect the rate or extent of absorption, or clearance of digoxin (0.5 mg followed by 0.25 mg daily) when administered in healthy adults. Digoxin did not affect the pharmacokinetic profile of tigecycline. Therefore, no dosage adjustment is necessary when tigecycline is administered with digoxin. In *in vitro* studies, no antagonism has been observed between tigecycline and other commonly used antibiotic classes.

Concurrent use of antibiotics with oral contraceptives may render oral contraceptives less effective.

Pregnancy and lactation

Pregnancy

Tigecycline may cause foetal harm when administered during pregnancy. The potential risk for humans is unknown. As it is known for tetracycline class antibiotics, tigecycline may also induce permanent dental defects (discoloration and enamel defects) and a delay in ossification processes in foetuses, exposed *in utero* during the last half of gestation, and in children under eight years of age due to the enrichment in tissues with a high calcium turnover and formation of calcium chelate complexes. Tigecycline should not be used during pregnancy unless clearly necessary.

Breast feeding

It is not known whether tigecycline/metabolites are excreted in human milk. Tigecycline is excreted into milk of lactating rats. Because a potential risk to the breastfeeding infant cannot be ruled out, when treating with tigecycline, caution should be exercised and interruption of breast-feeding should be considered.

Fertility

Tigecycline did not affect mating or fertility in rats at exposures up to 4.7 times the human daily dose based on AUC. In female rats, there were no compound-related effects on ovaries or oestrus cycles at exposures up to 4.7 times the human daily dose based on AUC.

Side effects

Frequency categories are expressed as: Very common (≥1/10); Common (≥1/100 to <1/10); Uncommon (≥1/1,000 to <1/100); Rare (≥1/10,000 to <1/1,000); Very rare (<1/10,000); Not known (cannot be estimated from the available data)

Infections and infestations:

Common: Sepsis/septic shock, pneumonia, abscess, infections

Blood and the lymphatic system disorders:

Common: Prolonged activated partial thromboplastin time (aPTT), Prolonged prothrombin time (PT)
Uncommon: Thrombocytopenia, Increased International Normalised Ratio (INR)

Immune system disorders:

Not known: Anaphylaxis/anaphylactoid reactions.

Metabolism and nutrition disorders:

Common: Hypoglycaemia, hypoproteinaemia

Nervous system disorders:

Common: Dizziness

Vascular disorders:

Common: Phlebitis Uncommon: Thrombophlebitis

Gastrointestinal disorders:

Very common: Nausea, vomiting, diarrhea
Common: Abdominal pain, dyspepsia, anorexia
Uncommon: Acute pancreatitis

Hepatobiliary disorders:

Common: Elevated aspartate aminotransferase (AST) in serum, and elevated alanine aminotransferase (ALT) in serum, hyperbilirubinaemia

Uncommon: Jaundice, liver injury, mostly cholestatic

Not known: Hepatic failure

Skin and subcutaneous tissue disorders:

Common: Pruritus, rash
Not known: Severe skin reactions, including Stevens - Johnson syndrome

General disorders and administration site conditions:

Common: Impaired healing, injection site reaction, headache

Uncommon: Injection site inflammation, injection site pain, injection site oedema, injection site phlebitis

Investigations:

Common: Elevated amylase in serum, increased blood urea nitrogen (BUN)

Description of selected adverse reactions

Antibiotic Class Effects:

Pseudomembranous colitis which may range in severity from mild to life threatening Overgrowth of non-susceptible organisms, including fungi

Tetracycline Class Effects:

Glycylcycline class antibiotics are structurally similar to tetracycline class antibiotics. Tetracycline class adverse reactions may include photosensitivity, pseudotumour cerebri, pancreatitis, and anti-anabolic action which has led to increased BUN, azotaemia, acidosis, and hyperphosphataemia.

Tigecycline may be associated with permanent tooth discolouration if used during tooth development

Symptoms and Treatment of over dose

No specific information is available on the treatment of over dosage. Intravenous administration of tigecycline at a single dose of 300 mg over 60 minutes in healthy volunteers resulted in an increased incidence of nausea and vomiting. Tigecycline is not removed in significant quantities by hemodialysis.

Storage condition

Storage condition for unopened vial: Store below 30°C. Store in the original package in order to protect from light.

Storages condition with dilution 0.9 % w/v Sodium Chloride

Store up to 24 hours (6 hours in the vial and the remaining time in the intravenous bag) at room temperature 25°C. If the storage conditions exceeds 25°C after reconstitution, tigecycline should be used immediately. Alternatively, TIGAR 50 mixed with 0.9% Sodium Chloride Injection, USP may be stored refrigerated at 2°C to 8°C for up to 48 hours following immediate transfer of the reconstituted solution into the intravenous bag.

5 % w/v Dextrose Injection

Store up to 24 hours (6 hours in the vial and the remaining time in the intravenous bag) at room temperature 25°C. If the storage conditions exceeds 25°C after reconstitution, tigecycline should be used immediately. Alternatively, TIGAR 50 mixed with 5 % w/v Dextrose Injection, USP may be stored refrigerated at 2°C to 8°C for up to 48 hours following immediate transfer of the reconstituted solution into the intravenous bag.

Lactated Ringer's solution

Store up to 24 hours (6 hours in the vial and the remaining time in the intravenous bag) at room temperature 25°C. If the storage conditions exceeds 25°C after reconstitution, tigecycline should be used immediately. Alternatively, TIGAR 50 mixed with Lactated Ringer's Solution may be stored refrigerated at 2°C to 8°C for up to 48 hours following immediate transfer of the reconstituted solution into the intravenous bag.

Shelf Life

Unopened vial: 2 years

After opening of the vial:

Each vial is for single use and should be used immediately after opening. If not used immediately, in-use storage times and conditions are the responsibility of the user.

Once added to the infusion bag:

From a microbiological point of view, reconstitution/dilution must take place in controlled and aseptic conditions and the medicinal product should be used immediately.

If not used immediately, in-use storage times and conditions are the responsibility of the user.

Store up to 24 hours (6 hours in the vial and the remaining time in the intravenous bag) at room temperature 25°C. This should be used within 24 hours.

Presentation

Sterile, lyophilized cake/powder for Injection in 5mL USP Type- I clear glass Lyo Vial. Each Vial is packed into secondary carton along with a package insert.

Manufactured by:



Gland Pharma Limited
Survey No. 143-148, 150 &151,
Near Gandimaisamma Cross Roads,
D.P. Pally, Dundigal Post,
Dundigal - Gandimaisamma Mandal,
Medchal – Malkajgiri District,
Hyderabad – 500043, Telangana, India

Product Registration Holder/Imported by:

UNIMED SDN BHD
53, Jalan Tembaga SD 5/2B, Bandar Sri Damansara,
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